Grade 9 Mathematics – Area

Answer Sheet – Worked Solutions

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Formulas: Parallelogram = base × height • Trapezium = ½(a + b)h • Circle = πr^2

Question 1

(i) Area = base × height = 14 × 6 = **84 cm^2**

(ii) Perimeter = 2(14 + 8) = 2 × 22 = **44 cm**

(iii) Area = BC × h → 84 = 8 × h → h = 84 ÷ 8 = **10.5 cm**

Question 2

(i) 96 = 12 × h → h = 96 ÷ 12 = **8 cm**

(ii) 2(PQ + QR) = 40 → PQ + QR = 20 → QR = 20 – 12 = **8 cm**

(iii) Area = QR × distance → 96 = 8 × d → d = **12 cm**

Question 3

(i) Area = ½(9 + 15) × 8 = ½ × 24 × 8 = **96 cm^2**

(ii) length × 8 = 96 → length = **12 cm**

(iii) New area = ½(9 + 18) × 8 = ½ × 27 × 8 = 108 cm^2; increase = 108 – 96 = **12 cm^2**

Question 4

(i) 70 = ½(6 + 8) × h → 70 = 7h → h = **10 cm**

(ii) 80 = ½(5 + b) × 10 → 80 = 5(5 + b) → 5 + b = 16 → b = **11 cm**

(iii) 70 = ½(a + b) × 10 → a + b = 14. Any two pairs, e.g. **4 cm and 10 cm**, **5 cm and 9 cm** (also 3 & 11, 2 & 12, 6 & 8, …)

Question 5

(i) Area = πr^2 = 22/7 × 7 × 7 = **154 cm^2**

(ii) r = 28 ÷ 2 = 14 cm; Area = 22/7 × 14 × 14 = **616 cm^2**

(iii) 22/7 × r^2 = 1386 → r^2 = 1386 × 7 ÷ 22 = 441 → r = **21 cm** (radius cannot be negative)

Question 6

(i) Area of square = 14 × 14 = **196 cm^2**

(ii) Diameter of circle = side of square = 14 cm, so radius = **7 cm**

(iii) Area of circle = 22/7 × 7 × 7 = **154 cm^2**

(iv) Remaining area = 196 – 154 = **42 cm^2**

Question 7

(i) Rectangle = 8 × 3 = **24 m^2**

(ii) Trapezium = ½(8 + 4) × 2 = **12 m^2**

(iii) Window = 22/7 × 0.7 × 0.7 = **1.54 m^2**

(iv) Area to paint = 24 + 12 – 1.54 = **34.46 m^2**

(v) Cost = 34.46 × 200 = **Rs. 6892**